Rakesh Dey

+91 7595848990 | @ id.rakeshdey@gmail.com, rakeshdey_t@isical.ac.in | LinkedIn | GitHub | Kolkata, India

Research Interest

I am passionate about comprehending the theoretical aspects of **Machine Learning**, with a specific focus on their application in health care technologies.

Education

Maulana Abul Kalam Azad University of Technology (WBUT): RCCIIT Kolkata, India
B.Tech. in Computer Science and Engineering; ***GPA: 8.77/10*** *August 2017 – July 2021*
Project: *Hand written digits classification using Convolutional Neural Network*

- **Relevant courses:** Artificial Intelligence (Introduction to Image Processing, Computer Vision), Mathematics I, II (Linear Algebra, Calculus, ODE, Probability, Statistics), Numerical Methods, Design & Analysis of Algorithm, Introductory Computer Graphics.

Skills

Programming Languages: Python, C/C++

Libraries: Pytorch, OpenCV, Scikit-Learn, Keras, TensorFlow, NumPy, Pandas, Matplotlib, dlib

Documentation & Typesetting: LaTeX, MS Office

Technologies: Git, Linux

Languages: English (Professional Proficiency), Bengali (Native), Hindi

Experience

Sikkim Manipal Institute of Technology Sikkim, India
Research *Feb 2026 – Present*

- Conducting research on lesion segmentation using machine learning algorithms to improve healthcare solutions.
  Supervisor: Prof. Palash Ghosal, Sikkim Manipal Institute of Technology, Sikkim

CVPR Unit, Indian Statistical Institute Kolkata, India
Project Linked Person *Oct 2022 – Feb 2026*

- Worked on the advancement of self-supervised contrastive learning by dynamically varying temperature.
  Project: Dynamic temperature scaling for self-supervised contrastive learning
  Collaborator & supervisor: Siladittya Manna, Indian Statistical Institute, Kolkata; Prof. Saumik Bhattacharya, Indian Institute of Technology, Kharagpur; Prof. Umapada Pal, Indian Statistical Institute, Kolkata
- Developed a novel method for estimating heart rate from facial video data.
  Project: Heart rate estimation from video
  Supervisor: Prof. Umapada Pal, CVPR Unit, Indian Statistical Institute, Kolkata; Prof. Palaiahnakote Shivakumara, University of Malaya, Kuala Lumpur, Malaysia
- Created and implemented a system that utilizes facial video input to accurately predict real-time heart rates.
  Project: Heart rate estimation from video
  Supervisor: Prof. Umapada Pal, CVPR Unit, Indian Statistical Institute, Kolkata; Prof. Palaiahnakote Shivakumara, University of Malaya, Kuala Lumpur, Malaysia

CVPR Unit, Indian Statistical Institute Kolkata, India
Research Internship *Apr 2022 – Oct 2022*

- Conducted a comprehensive survey on contemporary techniques for measuring blood oxygen saturation.
  Supervisor: Prof. Umapada Pal, CVPR Unit, Indian Statistical Institute, Kolkata
- Created a dataset specifically designed for the estimation of heart rate from facial video data.
  Supervisor: Prof. Umapada Pal, CVPR Unit, Indian Statistical Institute, Kolkata

Tata Consultancy Service Kolkata, India
Associate System Engineer Trainee *Sept 2021 – Apr 2022*

Research Projects

Lesion Segmentation — Sikkim, India
Working on lesion segmentation-related problems using computational algorithms. — *Feb 2026 – Ongoing*

Heart Rate Estimation System Design — Kolkata, India
Designed and implemented a real-time working system capable of estimating heart rate from facial video with high accuracy. — *Mar 2025 – Feb 2026*

Heart Rate Estimation from Facial Video V-2 — Kolkata, India
Examined and improved the accuracy of heart rate measurement with the help of spatial and frequency domain information sharing. — *Nov 2023 – Mar 2025*

Dynamic Temperature Scaling in Self-supervised Contrastive Learning — Kolkata, India
This project enhances self-supervised contrastive learning by introducing a novel cosine similarity-based dynamic temperature scaling for the InfoNCE loss. The proposed method adaptively balances attraction and repulsion between samples, improving feature distribution in the latent space. — *Mar 2023 – Nov 2023*

Explored StyleGAN Latent Space for Image Editing Task — Kolkata, India
This project proposes a latent optimization-based method for cross-domain image style transfer using the latent space of StyleGAN. By disentangling the latent space, the approach enables effective and robust image-to-image and text-to-image style transfer across diverse domains. — *Oct 2023 – Mar 2024*

Heart Rate Estimation from Facial Video — Kolkata, India
Examined the impact of spatial and frequency domain information on heart rate estimation. — *Dec 2022 – Oct 2023*

Survey on Blood Oxygen Saturation Measurement — Kolkata, India
Conducted a comprehensive study on blood oxygen saturation measurement methods and explored the potential of computer vision techniques for the same. — *Jan 2022 – Jun 2022*

Publications

- **Rakesh Dey**, Palaiahnakote Shivakumara, Saumik Bhattacharya, Umapada Pal, Sukalpa Chanda, "A New StyleGAN Latent Space Based Model for Image Style Transfer" (*Accepted in **ICPR - 2024***)
- Siladittya Manna, Soumitri Chottopadhay, **Rakesh Dey**, Saumik Bhattacharya, Umapada Pal, "DySTreSS: Dynamically Scaled Temperature in Self-Supervised Contrastive Learning" (*Accepted in **IEEE Transactions on Artificial Intelligence***, arxiv)
- **Rakesh Dey**, Palaiahnakote Shivakumara, Saumik Bhattacharya, Umapada Pal, Sukalpa Chanda, "SFTT: A Spatial-Frequency-Temporal based End-to-End Transformer for Heart Rate Estimation" (*Accepted in **IEEE Transactions on Emerging Topics in Computational Intelligence***)

Awards, Achievements & Certificates

Scholarship: Received "Half Fee Scholarship for Excellent Academic Records" for four consecutive years (2017 - 2021).
Achievements: Secured 2^{nd} & 1^{st} position in college & school art competitions respectively; elected as the president of Ramakrishna Mission Rahara school mathematics club.
Certificates: Received certificate of excellence for an outstanding summer project from GLOBSYN BUSINESS SCHOOL (2018).

Volunteer Services

- Peer reviewer: **Springer Nature Computer Science**, **ICASSP 2025**
- Volunteering experience: International Conference on Pattern Recognition **(ICPR)** 2024
- Invited talk: Gave a talk on **Large Language Models** in the Faculty Development Program at **Sikkim Manipal Institute of Technology, India**.
- Demo presentation: **Deep Learning for Physiological Measurements**
  Gave a small talk on Deep Learning techniques for estimating physiological parameters and presented a demo for estimating heart rate from facial video.